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Subject: Digital Signal Processing

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Causal versus noncausal systems. We begin with the definition of causal discrete-time systems.

Definition. A system is said to be *causal* if the output of the system at any time n [i.e., $y(n)$] depends only on present and past inputs [i.e., $x(n)$, $x(n - 1)$, $x(n - 2)$, ...], but does not depend on future inputs [i.e., $x(n + 1)$, $x(n + 2)$, ...]. In mathematical terms, the output of a causal system satisfies an equation of the form

$$y(n) = F[x(n), x(n - 1), x(n - 2), \dots] \quad (2.2.44)$$

where $F[\cdot]$ is some arbitrary function.

If a system does not satisfy this definition, it is called *noncausal*. Such a system has an output that depends not only on present and past inputs but also on future inputs.

It is apparent that in real-time signal processing applications we cannot observe future values of the signal, and hence a noncausal system is physically unrealizable (i.e., it cannot be implemented). On the other hand, if the signal is recorded so that the processing is done off-line (nonreal time), it is possible to implement a noncausal system, since all values of the signal are available at the time of processing. This is often the case in the processing of geophysical signals and images.

Example 2.2.6

Determine if the systems described by the following input–output equations are causal or noncausal.

$$\text{(a)} \quad y(n) = x(n) - x(n - 1) \quad \text{(b)} \quad y(n) = \sum_{k=-\infty}^n x(k) \quad \text{(c)} \quad y(n) = ax(n)$$

$$\text{(d)} \quad y(n) = x(n) + 3x(n + 4) \quad \text{(e)} \quad y(n) = x(n^2) \quad \text{(f)} \quad y(n) = x(2n)$$

$$\text{(g)} \quad y(n) = x(-n)$$

Solution The systems described in parts (a), (b), and (c) are clearly causal, since the output depends only on the present and past inputs. On the other hand, the systems in parts (d), (e), and (f) are clearly noncausal, since the output depends on future values of the input. The system in (g) is also noncausal, as we note by selecting, for example, $n = -1$, which yields $y(-1) = x(1)$. Thus the output at $n = -1$ depends on the input at $n = 1$, which is two units of time into the future.

Stable versus unstable systems. Stability is an important property that must be considered in any practical application of a system. Unstable systems usually exhibit erratic and extreme behavior and cause overflow in any practical implementation. Here, we define mathematically what we mean by a stable system, and later, in Section 2.3.6, we explore the implications of this definition for linear, time-invariant systems.

Definition. An arbitrary relaxed system is said to be bounded input–bounded output (BIBO) stable if and only if every bounded input produces a bounded output.

The conditions that the input sequence $x(n)$ and the output sequence $y(n)$ are bounded is translated mathematically to mean that there exist some finite numbers, say M_x and M_y , such that

$$|x(n)| \leq M_x < \infty \quad |y(n)| \leq M_y < \infty \quad (2.2.45)$$

for all n . If, for some bounded input sequence $x(n)$, the output is unbounded (infinite), the system is classified as unstable.

Example 2.2.7

Consider the nonlinear system described by the input–output equation

$$y(n) = y^2(n-1) + x(n)$$

As an input sequence we select the bounded signal

$$x(n) = C\delta(n)$$

where C is a constant. We also assume that $y(-1) = 0$. Then the output sequence is

$$y(0) = C, \quad y(1) = C^2, \quad y(2) = C^4, \quad \dots, \quad y(n) = C^{2^n}$$

Clearly, the output is unbounded when $1 < |C| < \infty$. Therefore, the system is BIBO unstable, since a bounded input sequence has resulted in an unbounded output.

2.3 ANALYSIS OF DISCRETE-TIME LINEAR TIME-INVARIANT SYSTEMS

2.3.2 Resolution of a Discrete-Time Signal into Impulses

Suppose we have an arbitrary signal $x(n)$ that we wish to resolve into a sum of unit sample sequences. To utilize the notation established in the preceding section, we select the elementary signals $x_k(n)$ to be

$$x_k(n) = \delta(n - k) \quad (2.3.1)$$

where k represents the delay of the unit sample sequence. To handle an arbitrary

Now suppose that we multiply the two sequences $x(n)$ and $\delta(n - k)$. Since $\delta(n - k)$ is zero everywhere except at $n = k$, where its value is unity, the result of this multiplication is another sequence that is zero everywhere except at $n = k$, where its value is $x(k)$, as illustrated in Fig. 2.22. Thus

$$x(n)\delta(n - k) = x(k)\delta(n - k) \quad (2.3.8)$$

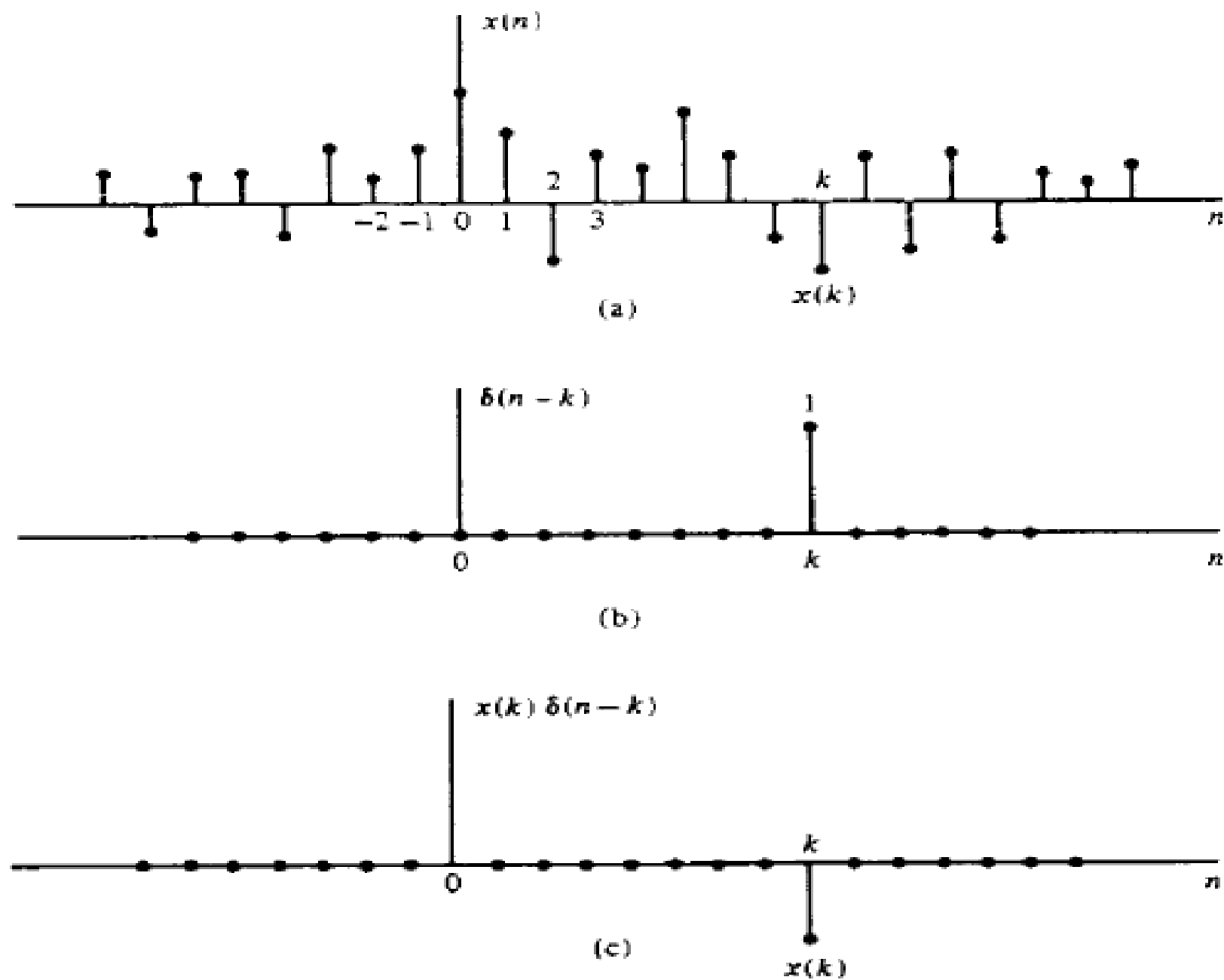


Figure 2.22 Multiplication of a signal $x[n]$ with a shifted unit sample sequence.

In other words, each multiplication of the signal $x(n)$ by a unit impulse at some delay k , [i.e., $\delta(n - k)$], in essence picks out the single value $x(k)$ of the signal $x(n)$ at the delay where the unit impulse is nonzero. Consequently, if we repeat this multiplication over all possible delays, $-\infty < k < \infty$, and sum all the product sequences, the result will be a sequence equal to the sequence $x(n)$, that is,

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n - k) \quad (2.3.10)$$

Example 2.3.1

Consider the special case of a finite-duration sequence given as

$$x(n) = \{2, 4, 0, 3\}$$

↑

Resolve the sequence $x(n)$ into a sum of weighted impulse sequences.

Solution Since the sequence $x(n)$ is nonzero for the time instants $n = -1, 0, 2$, we need three impulses at delays $k = -1, 0, 2$. Following (2.3.10) we find that

$$x(n) = 2\delta(n + 1) + 4\delta(n) + 3\delta(n - 2)$$

2.3.3 Response of LTI Systems to Arbitrary Inputs: The Convolution Sum

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) \quad (2.3.17)$$

The formula in (2.3.17) that gives the response $y(n)$ of the LTI system as a function of the input signal $x(n)$ and the unit sample (impulse) response $h(n)$ is called a *convolution sum*. We say that the input $x(n)$ is convolved with the impulse

To summarize, the process of computing the convolution between $x(k)$ and $h(k)$ involves the following four steps.

1. *Folding.* Fold $h(k)$ about $k = 0$ to obtain $h(-k)$.
2. *Shifting.* Shift $h(-k)$ by n_0 to the right (left) if n_0 is positive (negative), to obtain $h(n_0 - k)$.
3. *Multiplication.* Multiply $x(k)$ by $h(n_0 - k)$ to obtain the product sequence $v_{n_0}(k) \equiv x(k)h(n_0 - k)$.
4. *Summation.* Sum all the values of the product sequence $v_{n_0}(k)$ to obtain the value of the output at time $n = n_0$.

Example 2.3.2

The impulse response of a linear time-invariant system is

$$h(n) = \{1, 2, 1, -1\} \quad (2.3.19)$$

↑

Determine the response of the system to the input signal

$$x(n) = \{1, 2, 3, 1\} \quad (2.3.20)$$

↑

Solution We shall compute the convolution according to the formula (2.3.17), but we shall use graphs of the sequences to aid us in the computation. In Fig. 2.23a we illustrate the input signal sequence $x(k)$ and the impulse response $h(k)$ of the system, using k as the time index in order to be consistent with (2.3.17).

The first step in the computation of the convolution sum is to fold $h(k)$. The folded sequence $h(-k)$ is illustrated in Fig. 2.23b. Now we can compute the output at $n = 0$, according to (2.3.17), which is

$$y(0) = \sum_{k=-\infty}^{\infty} x(k)h(-k) \quad (2.3.21)$$

Since the shift $n = 0$, we use $h(-k)$ directly without shifting it. The product sequence

$$v_0(k) \equiv x(k)h(-k) \quad (2.3.22)$$

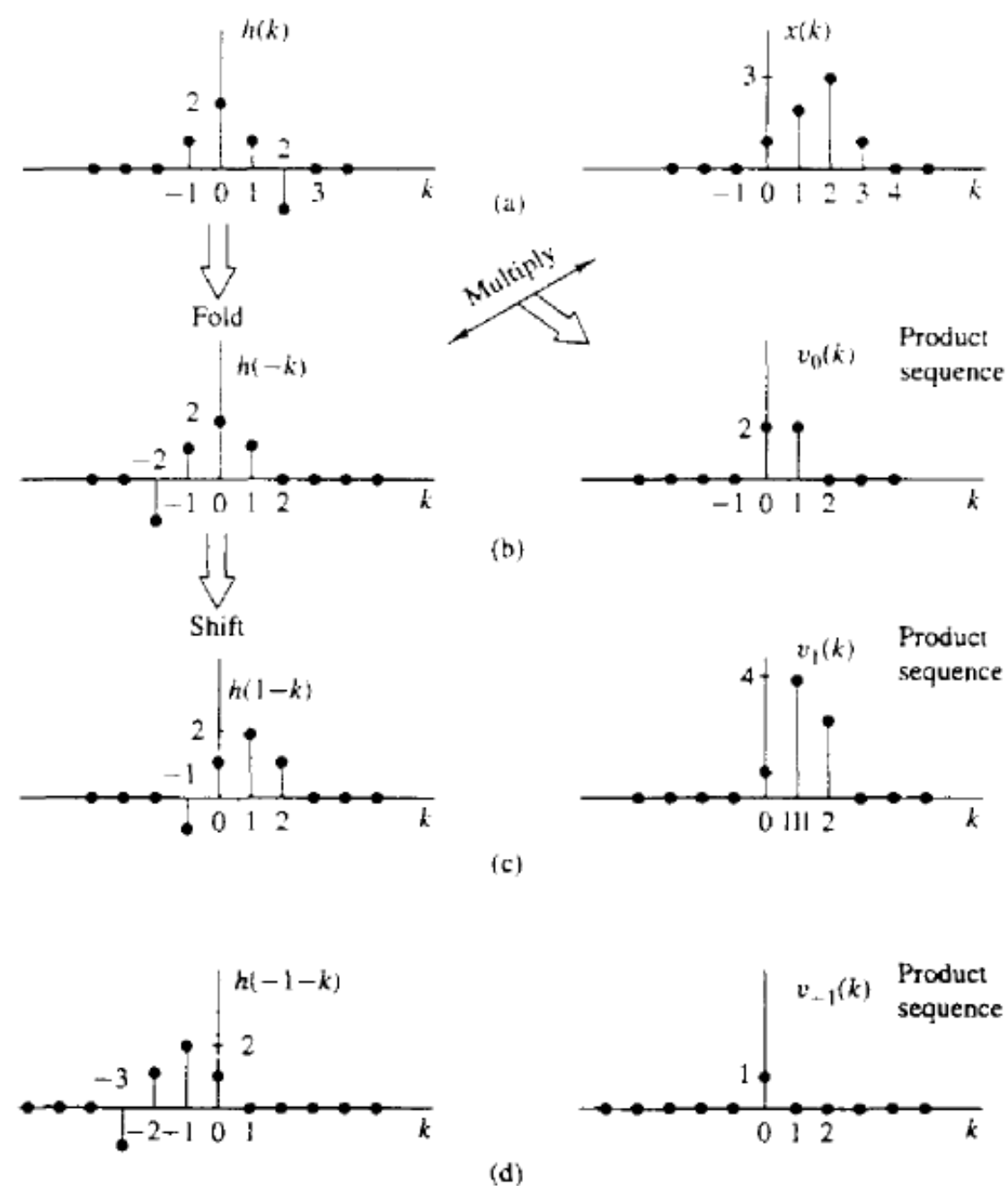


Figure 2.23 Graphical computation of convolution.

$$v_0(k) \equiv x(k)h(-k) \quad (2.3.22)$$

is also shown in Fig. 2.23b. Finally, the sum of all the terms in the product sequence yields

$$y(0) = \sum_{k=-\infty}^{\infty} v_0(k) = 4$$

We continue the computation by evaluating the response of the system at $n = 1$. According to (2.3.17),

$$y(1) = \sum_{k=-\infty}^{\infty} x(k)h(1-k) \quad (2.3.23)$$

The sequence $h(1-k)$ is simply the folded sequence $h(-k)$ shifted to the right by one unit in time. This sequence is illustrated in Fig. 2.23c. The product sequence

$$v_1(k) \equiv x(k)h(1-k) \quad (2.3.24)$$

is also illustrated in Fig. 2.23c. Finally, the sum of all the values in the product sequence yields

$$y(1) = \sum_{k=-\infty}^{\infty} v_1(k) = 8$$

In a similar manner, we obtain $y(2)$ by shifting $h(-k)$ two units to the right, forming the product sequence $v_2(k) = x(k)h(2 - k)$ and then summing all the terms in the product sequence obtaining $y(2) = 8$. By shifting $h(-k)$ farther to the right, multiplying the corresponding sequence, and summing over all the values of the resulting product sequences, we obtain $y(3) = 3$, $y(4) = -2$, $y(5) = -1$. For $n > 5$, we find that $y(n) = 0$ because the product sequences contain all zeros. Thus we have obtained the response $y(n)$ for $n > 0$.

Next we wish to evaluate $y(n)$ for $n < 0$. We begin with $n = -1$. Then

$$y(-1) = \sum_{k=-\infty}^{\infty} x(k)h(-1-k) \quad (2.3.25)$$

Now the sequence $h(-1-k)$ is simply the folded sequence $h(-k)$ shifted one time unit to the left. The resulting sequence is illustrated in Fig. 2.23d. The corresponding product sequence is also shown in Fig. 2.23d. Finally, summing over the values of the product sequence, we obtain

$$y(-1) = 1$$

From observation of the graphs of Fig. 2.23, it is clear that any further shifts of $h(-1 - k)$ to the left always results in an all-zero product sequence, and hence

$$y(n) = 0 \quad \text{for } n \leq -2$$

Now we have the entire response of the system for $-\infty < n < \infty$, which we summarize below as

$$y(n) = \{ \dots, 0, 0, 1, \underset{\uparrow}{4}, 8, 8, 3, -2, -1, 0, 0, \dots \} \quad (2.3.26)$$